

Technology Stack

Date	15 March 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Technology Stack:

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (HTML/CSS)	<i>HTML5, CSS3, Bootstrap</i>	<i>Used to create a responsive, user-friendly web interface for entering HDI indicators and displaying prediction results. Bootstrap ensures a mobile-friendly and attractive design.</i>

2	Backend / Server-Side (Python/Django, Java)	Python, Flask	Python is widely used for Machine Learning. Flask is a lightweight web framework that integrates easily with ML models and processes user requests efficiently.
3	Database / Data Storage (MySQL, MongoDB, PostgreSQL)	CSV Dataset, Pandas DataFrames	The HDI dataset is stored in CSV format and processed using Pandas. It is sufficient for machine learning model training and prediction without requiring a database server.
4	Cloud / Hosting / Deployment (AWS, Azure, Heroku, Docker)	Local Flask Server / Render / PythonAnywhere	Used to deploy the Flask application so users can access the HDI prediction system through a web browser. These platforms are easy to deploy and cost-effective for student projects.
5	Version Control & CI/CD (GitHub, GitLab, Jenkins)	[Specify technology]	[Explain why this was chosen (e.g., seamless collaboration, automated testing)]

6	<i>Third-Party APIs / Other Tools</i> <i>(Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Joblib, Jupyter Notebook, Anaconda</i>	<i>Scikit-learn is used for building the ML model, Pandas and NumPy for data processing, Matplotlib and Seaborn for visualization, Joblib for saving the trained model, and Jupyter Notebook/Anaconda for development and testing.</i>